

Agenda:

(5 min)

✓ → why joins might be needed

✓ → alternatives

✓ → inner join ~→ [D.F. working of joins]

✓ → self joins, compound join

→ left and right outer joins

→ full outer join

→ cross join.

→ Hash join and merge join]
sort

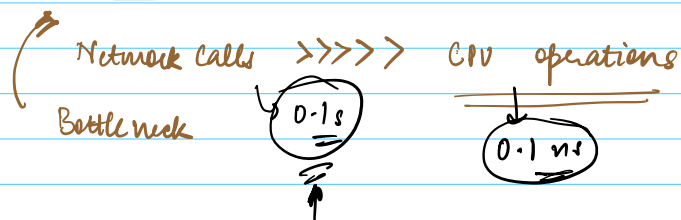
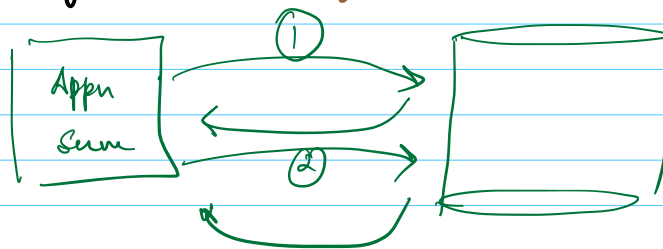
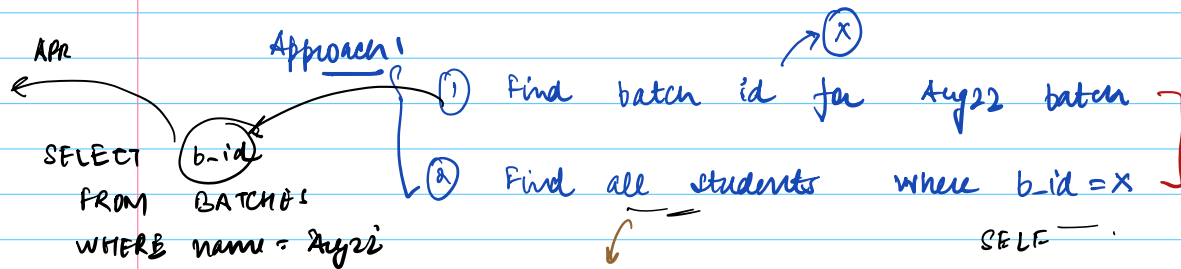
All queries queried a single table.

FROM <TABLE NAME>

↳ only 1 table?

Students			batches	
id	name	b_id	id	name
1	ABC	1	1	Aug 22
2	XYZ	2	2	Sept 22
3	CDE	1		

→ Find all the students in 'Aug 22' batch



Approach-2

change the DB schema

id	name	b_name

Redundancy
↓
Problem

Approach-3 → Joins



No redundancy

No network

SQL performs internal optimisation
to make joins faster

students			batches	
id	name	b_id	id	name
1	ABC	1	1	Aug 22
2	XYZ	2	2	Sept 22
3	CDE	1		

Students - batches

id	name	b_id	id	name
1	ABC	1	1	Aug 22
2	XYZ	2	2	Sept 22
3	CDE	1	1	Aug 22

Joins help us with virtually merging tables for the purpose of query

NOT ACTUALLY CREATE A TABLE IN DB

Join table A and B

↓
specify which column is the basis of join

A and B on col X

ans = []

for row₁ in A : ↩

for row₂ in B :

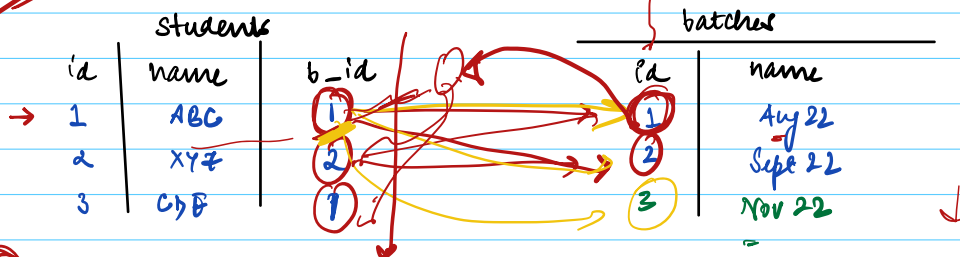
if row₁.X == row₂.X :

ans.add(concat(row₁, row₂))

CROSS JOIN

EXPLAIN

f



1 → (1)

A = students B = batches
X = b_id

(1, ABC, 1, 1, Aug22), (2, XYZ, 2, 2, Sep22)
(3, CDE, 1, 1, Aug22)

⊕ JOIN
== INNER JOIN

INNER JOIN: (most common)

Join on the basis of common column

A.X there is no value in B.X
that matches



Record will not show up in result

> find all the students in 'Aug22' batch



SQL=?

```
SELECT stu_id, name
```

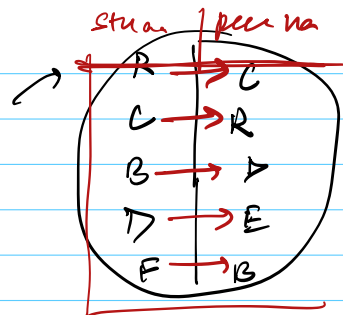
```
FROM STUDENTS INNER JOIN BATCHES
```

```
ON STUDENTS.b_id = BATCHES.id
```

```
WHERE BATCHES.name = 'Aug22'
```

Student Buddies

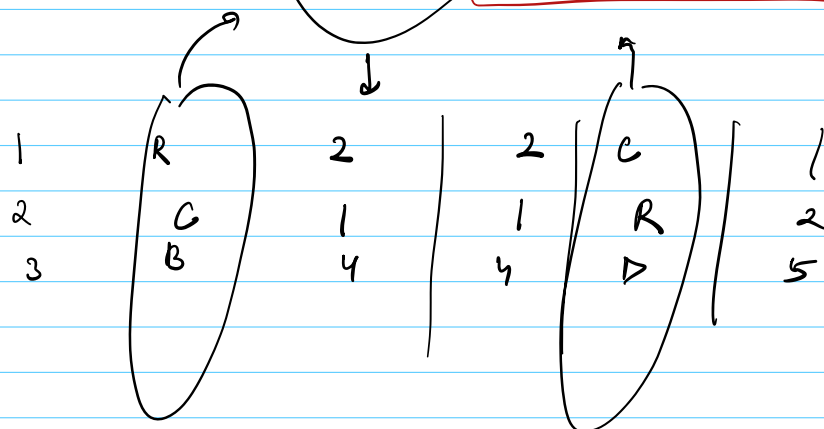
Students		
id	name	peer_id
1	R	2
2	C	1
3	B	4
4	D	5
5	E	3



Write a query that finds name → peer mapping

Students		
id	name	peer_id
1	R	2
2	C	1
3	B	4
4	D	5
5	E	3

Students		
id	name	peer_id
1	R	2
2	C	1
3	B	4
4	D	5
5	E	3



SELF JOIN:

```
SELECT  s.name, p.name  
FROM    students s INNER JOIN peer p  
ON      s.peer_id = p.id
```

↓

→ students

Different Aliases to the same table

A

are mandatory?

YES

id	name	b_id	<u>i_id</u>
----	------	------	-------------

MULTI TABLE JOINS

Students			batches			instructors	
id	name	b_id	id	b_name	i_id	id	name

s.name	i.name
--------	--------

```

SELECT    s.name    , i.name
FROM      STUDENT s  JOIN  BATCHES b
ON        s.b_id = b.id
JOIN      INSTRUCTORS i ON b.i_id = i.id;

```

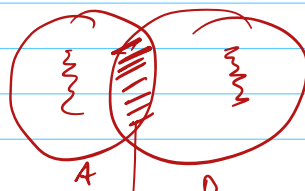
COMPOUND JOINS

T1				T2			
id	a	b	c	id	aa	b b	dd

ON $(T1.a = T2.aa \text{ AND } T1.b = T2.bb)$

USING KEY WORD

NATURAL JOIN



Inner Join

OUTER JOINS

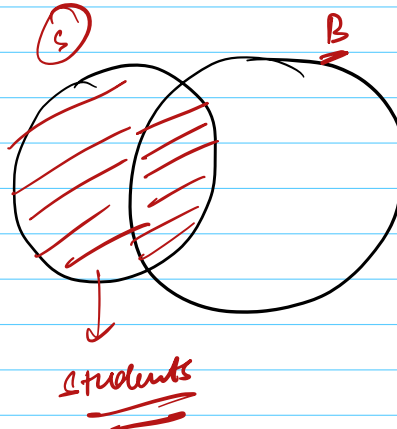
Students			Batches	
id	name	b_id	id	name
1	A	1	1	X
2	B	2	2	Y
3	C		3	Z

Students with no batch assigned → 0 students

Q1 ⇒ List all students in school with batch name, if no batch name → Null

1	A	X
2	B	Y

not be displayed in inner join



LEFT OUTER JOIN

```
-- [ SELECT s.name, b.name
    FROM STUDENTS s LEFT OUTER
    JOIN BATCHES b ON
    s.bid = b.id ]
```

Right outer join !

↳ Right side Table

All entries from right side will be present.

RIGHT OUTER JOIN

stu			Rater	
id	name	bid	id	name
1	A	1	1	X ✓
2	B	1	2	X
3	C	2	3	Y
4	D	3	4	Z
5	D	NULL		

1	A	W
2	B	W
3	C	X
4	D	Y
NU	NU	Z

Outer joins TRY to satisfy the condition mentioned. If for some record no record in other table matches it return that record matched with NULL inside

CROSS JOIN

id	name	bid		b_id	name
1	A	1	X	1	X
2	B	1		2	Y
3	C	2		3	Z

9 records in cross join

Cross join

3x3

1	A	1	b_id	X
1	A	1	2	Y
1	A	1	3	Z
2	B	1	1	X

left outer join

id	name	bid		b_id	name
1	A	1	X	1	X
2	B	1		2	Y
3	C	2		3	Z
4	D	NULL			

1	A	1		X
2	B	1		X
3	C	2	2	Y
4	D	NU	NU	NU

Right outer join

id	name	bid		b_id	name
1	A	1		1	X
2	B	1	X	2	Y
3	C	2		3	Z
4	D	NULL			

1	A	1	1	X
2	B	1	1	X
3	C	2	2	Y
NULL	NULL	NULL	3	Z

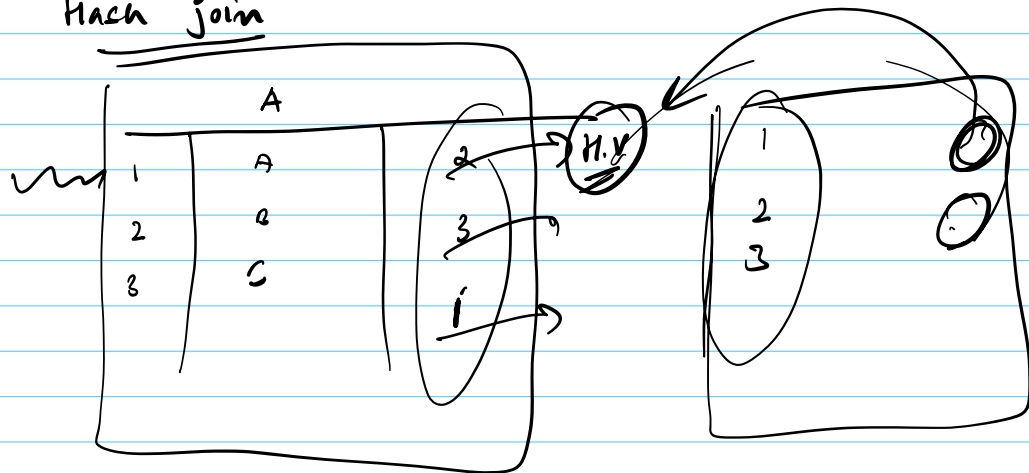
id	name	bid		b_id	name
1	A	1		1	X
2	B	1	X	2	Y
3	C	2		3	Z
4	D	NULL			

Full outer join

1	A	1	1	X
2	B	1	1	X
3	C	2	2	Y
4	D	NULL	NULL	NULL
NULL	NULL	NULL	3	Z

MySQL doesn't support full outer join

Hash join



$$O(N + M)$$

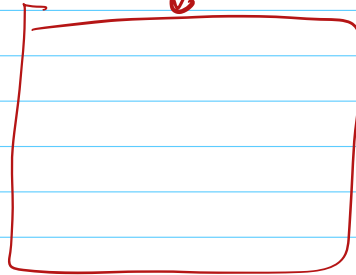
because of hash maps

Merge Joins
sort

↳ External merge sort

ON -id = id
AND mark > 90
↓
sorted by mark

STUDENTS, BATCHES



CROSS JOIN
of both
tables

NAM